



# The Chemistry a Culture Performs

Teacher copy (everything in the student copy, plus answers and teaching notes)

## The article, marked for rhetorical devices (one color throughout):

1 A crock of cabbage sits on a kitchen counter, salted, weighted, sealed away from the air. Nothing about it looks busy. Yet inside, a population that cannot be seen is already at work, breaking glucose without oxygen and converting it, molecule by molecule, into the acid that will keep the harvest from rotting. The cabbage does not change because the cook willed it to. It changes because something living is metabolizing inside it. [Personification]

2 What that population is doing has a name and a chemistry. Fermentation is anaerobic respiration: microorganisms pull energy from glucose with no oxygen present, and they leave behind organic end products, lactic acid in one pathway, ethanol and carbon dioxide in another. The bacterium most responsible for the souring crock, *Lactobacillus*, runs the first route. One molecule of glucose becomes two molecules of lactic acid, by way of glycolysis and an enzyme called lactate dehydrogenase.

3 That enzyme does not improvise. It catalyzes one reaction, the reduction of pyruvate to lactate, accepting electrons from a carrier called NADH, and in doing so it hands back the NAD<sup>+</sup> that glycolysis needs to keep running; without that handoff the whole sequence would stall after a single pass. [Cumulative and periodic sentences] The precision is almost hard to credit: lactate dehydrogenase places its atoms with an estimated error rate below one in a billion, so only one optical form of the acid, the D or the L, comes out, depending on which version of the enzyme the microbe carries. [Logos]

4 Yeast performs a different chemistry on the same sugar. *Saccharomyces cerevisiae*, the organism in bread and beer, splits its work across two enzymes: pyruvate decarboxylase strips a carbon away as carbon dioxide and leaves acetaldehyde, then alcohol dehydrogenase reduces that acetaldehyde to ethanol. The first enzyme needs a scrap of vitamin B1 and a magnesium ion to function. The second holds a zinc atom at its core and passes a hydride from NADH onto each molecule it converts. Both steps exist to do what the bacterium's single step also did: regenerate NAD<sup>+</sup>, so glycolysis can keep yielding its meager two molecules of ATP per glucose.

5 The conditions for all this are narrow, and the cook controls them more than the microbe does. Lactic acid bacteria grow best between 30 and 40 degrees Celsius, in a mild acidity that drifts as they work. Yeast prefers it cooler, 25 to 30 degrees, and turns sluggish and stressed near 35, its ethanol yield falling off. A few degrees is not a detail. It is the difference between the culture a cook wanted and a different one entirely.

6 A starter culture is the one deliberate intervention. Drop a defined population of selected strains into the substrate and they outcompete whatever wandered in, claim the sugar first, drop the pH, and in some cases secrete bacteriocins, small antimicrobial peptides that suppress the organisms a cook does not want.

[Asyndeton] What follows is not static. The community shifts in sequence: one species dominates, alters the

chemistry, and yields the lead to the next as the acid climbs; and in those dense crowded populations, cell pressed against cell, reactions occur that never occur in a microbe kept alone. [Imagery]

7 None of it is simple, and the same culture is not locked to one product. [Antithesis] Lactic acid bacteria can route pyruvate several ways depending on what surrounds them, toward lactate, toward acetyl-CoA and carbon dioxide, toward compounds that register on the tongue as flavor. The leuconostocs and weissellas take yet another path, the phosphoketolase route, and yield lactic acid, ethanol, and gas all at once. A cook salts a vegetable and waits. The cook supplies the temperature, the salt, the time, and the harvest. Everything else, the part that actually does the transformation, belongs to something with no intention at all, only a chemistry it cannot help but perform.

## Devices, and why they are here:

**Personification** (paragraph 1): “The cabbage does not change because the cook willed it to. It changes because something living is metabolizing inside it.”

Holding the change against the cook's will and crediting it to the living culture animates the microbe as the true agent, which is the whole angle: the human sets conditions, the organism does the work.

**Cumulative and periodic sentences** (paragraph 3): “That enzyme does not improvise. It catalyzes one reaction, the reduction of pyruvate to lactate, accepting electrons from a carrier called NADH, and in doing so it hands back the NAD<sup>+</sup> that glycolysis needs to keep running; without that handoff the whole sequence would stall after a single pass.”

A loose main clause that keeps adding coordinate detail enacts the relay of the reaction itself, letting the chemistry accumulate the way the spec wants wonder built, by piling up accurate particulars rather than asserting awe.

**Logos** (paragraph 3): “The precision is almost hard to credit: lactate dehydrogenase places its atoms with an estimated error rate below one in a billion, so only one optical form of the acid, the D or the L, comes out, depending on which version of the enzyme the microbe carries.”

The sub-billion error figure and the D/L isomer outcome reason from measured fact to the claim of precision, earning belief through detail instead of an adjective like astonishing.

**Asyndeton** (paragraph 6): “A starter culture is the one deliberate intervention. Drop a defined population of selected strains into the substrate and they outcompete whatever wandered in, claim the sugar first, drop the pH, and in some cases secrete bacteriocins, small antimicrobial peptides that suppress the organisms a cook does not want.”

Dropping conjunctions across outcompete, claim, drop, and secrete speeds the rhythm so the starter culture's takeover reads as the rapid cascade it is.

**Imagery** (paragraph 6): “The community shifts in sequence: one species dominates, alters the chemistry, and yields the lead to the next as the acid climbs; and in those dense crowded populations, cell pressed against cell, reactions occur that never occur in a microbe kept alone.”

Cell pressed against cell makes the abstract idea of population density concrete and physical, so the reader can picture the crowding that produces reactions an isolated cell never shows.

**Antithesis** (paragraph 7): “None of it is simple, and the same culture is not locked to one product.” Setting not simple against the single-product expectation, and one culture against many possible products, frames the paragraph's turn toward the microbe's metabolic range, the register's primary structural device.

**AP-style multiple choice:**

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1. In the fourth paragraph, the writer details yeast's two-enzyme pathway and then notes that both steps exist to regenerate NAD<sup>+</sup> "so glycolysis can keep yielding its meager two molecules of ATP per glucose." This detail primarily serves to

- A) argue that ethanol fermentation is more efficient than lactic acid fermentation
- B) clarify that yeast and bacteria, despite different end products, perform the same underlying chemical service
- C) establish that yeast requires more cofactors than bacteria and is therefore more advanced
- D) suggest that the energy yield of fermentation is too small to matter biologically

2. Which of the following best characterizes the writer's tone toward the microorganisms across the opener?

- A) celebratory enthusiasm that foregrounds the writer's own wonder
- B) cautious alarm at the risks of uncontrolled microbial activity
- C) detached skepticism about claims made for fermentation
- D) measured admiration grounded in technical precision rather than overt praise

**Multiple choice answer key:**

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1. **Answer B.** The passage explicitly says both yeast steps exist to do "what the bacterium's single step also did: regenerate NAD<sup>+</sup>," making the shared chemical purpose the point of the detail. The efficiency-comparison option fails as right-effect-wrong-function: the passage compares pathways but never ranks their efficiency, and "meager" applies to both. The more-advanced option fails as out-of-passage import and over-broad extreme; the text notes yeast's extra cofactors but draws no hierarchy of advancement. The too-small-to-matter option misreads the literal word "meager" as a claim that the yield is biologically negligible, when the passage treats that small yield as the reason the whole cycle runs.

2. **Answer D.** The tone is built from exact mechanism (error rates, named enzymes, temperature bands) and lands on respect without adjectives like amazing, which is admiration carried by precision. The skepticism option inverts the stance; the writer credits the microbes rather than doubting them. The celebratory-enthusiasm option overstates and misplaces the source: the writer recedes and never foregrounds personal wonder, so it fails by over-broad wording and wrong location of feeling. The alarm option imports a threat frame the passage never raises; spoilage is mentioned only as the thing fermentation prevents, not as a danger the writer warns against.

**Discussion questions:**

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1. The opener repeatedly frames the cook and the microbe as partners with divided labor: the cook supplies salt, temperature, and time, while the microbe supplies the chemistry. Where in the text is this division clearest, and how does the writer keep it from tipping into either pure human control or pure microbial autonomy?

2. The Long Look register builds wonder through accumulated precise detail rather than through words like astonishing. Find two moments where a technical particular (a number, an enzyme, a temperature) carries more awe than an adjective could, and explain how the precision itself does the emotional work.

**Sample strong discussion responses:**

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1. The division is clearest in the final paragraph, where the writer lists what the cook provides, the temperature, the salt, the time, and the harvest, and then assigns everything else to something with no intention at all. The balance holds because the writer keeps crediting the microbe with the actual transformation while insisting the conditions are human-set; the fifth paragraph makes this explicit by saying the cook controls the conditions more than the microbe does, yet a few degrees decides which culture results. Neither party is sovereign: the cook cannot perform the chemistry, and the microbe cannot choose its temperature.

2. The sub-billion error rate of lactate dehydrogenase in the third paragraph does what no adjective could: the smallness of one in a billion makes the enzyme's accuracy felt as a fact rather than a claim, and the consequence, that only one optical form of the acid emerges, gives the number a visible result. The temperature bands in the fifth paragraph work the same way; learning that yeast turns sluggish near 35 degrees and that a few degrees separates one culture from another makes the precision tangible. In both cases the awe comes from how exact the world turns out to be, not from the writer telling us to be impressed.

### **Writing prompt (Homework or extended in-class, Q2 rhetorical analysis):**

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The opener describes fermentation as a process that happens out of sight, framing the relationship between a cook and the living culture that transforms a harvest. Read the passage carefully. Then write an essay in which you analyze the rhetorical choices the writer makes to convey wonder at the chemistry of fermentation while withholding overt expressions of awe. Support your analysis with specific evidence from the text, and explain how the writer's choices of detail, sentence structure, and figurative language create the intended effect on the reader.

### **Common misconceptions to watch for:**

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- Students may read the opening personification (the cabbage changes because something living is metabolizing inside it) as a claim that microbes act with intention or purpose. The text deliberately ends on the opposite point: the microbe has no intention at all, only a chemistry it cannot help but perform. Personification here dramatizes agency without asserting a mind.
- Students often assume fermentation produces a large amount of energy because it transforms food so dramatically. The passage is explicit that glycolysis yields only two molecules of ATP per glucose, calling the yield meager; the visible transformation comes from the acid and gas, not from a big energy payoff.
- Students may treat lactic acid fermentation and ethanol fermentation as fundamentally different processes. The opener shows they share a core purpose, regenerating  $\text{NAD}^+$  so glycolysis can continue, and differ mainly in their end products and the number of enzyme steps.
- Students may think a starter culture is a single fixed organism doing one job. The sixth paragraph describes an ordered succession in which species hand off dominance as the acid climbs, and the seventh notes that even one culture can route pyruvate toward several different products depending on conditions.

### **AP exam alignment:**

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This opener teaches students to analyze how a writer in an observational register creates effect through precise technical detail, varied sentence length, and restrained figurative language rather than through emotive assertion (Style, Skill Category 7). The MCQs pair a reasoning-and-organization item on the function of a specific detail (Skill Category 5) with a style item on tone (Skill Category 7), and the Q2 prompt asks students to analyze the rhetorical choices that produce wonder without overt awe.